

Gradient Symbolic Representation and the [voice] specification of nasals in Japanese

THE EMPIRICAL PROBLEM: The [voice] feature specification of nasal consonants in Japanese presents an apparent paradox, which various phonologists have attempted to address in the past (Borowsky 1986; Ito & Mester 1986; Ito et al. 1995; Rice 1993, among several others). On the one hand, nasals trigger voicing of following voiceless consonants (e.g., /kam-ta/ → [kan-da] ‘bit’), which seems to suggest that nasals are specified for [+voice]. An analysis of half rhymes in Japanese rap lyrics likewise shows that nasals are more likely to rhyme with voiced obstruents than with voiceless obstruents, again suggesting that nasals are [+voice] (Kawahara 2007). On the other hand, Lyman’s Law appears to be insensitive to this [+voice] of nasals. More specifically, Rendaku turns the initial consonant of a second member of a compound to voiced (e.g., /hosi-sora/ → [hosi-zora] ‘starry sky’), but Lyman’s Law blocks this process if there is a voiced consonant in the second member already (e.g., /hosi-saba/ → [hosi-saba], *[hosi-zaba] ‘dried mackerel’). However, nasals do not block Rendaku, i.e., it does not trigger Lyman’s Law (e.g., /neko-suna/ → /neko-zuna/ ‘cat sand’), which appears to suggest that nasals do not bear the [+voice] feature.

Recent studies have revealed an extra layer of puzzle to this apparent paradox: Rendaku application is unaffected by one instance of nasal, but its application probability seems to be reduced by two instances of nasals. This new observation has been argued to hold based on a corpus study (Kim 2019) using the Corpus of the Spontaneous Japanese (Maekawa et al. 2004). Kim (2019) argues that items that contain two nasal consonants never undergo Rendaku in the whole corpus (e.g., /hanami/ ‘cherry-watching’ and /kaname/ ‘pivot’). The same tendency was observed in an experimental study using nonce words (Kumagai 2017), which shows that a sequence of two nasals reduced the Rendaku application rate by about 10% compared to the baseline. A reanalysis of this result using Bayesian mixed effects logistic regression shows that having two nasals meaningfully reduced the Rendaku application rate ($\beta_1 = -0.49$, 95% Credible Interval = [-0.82, -0.18]).

THE PROPOSAL: This paper proposes a solution to this apparently puzzling behavior of Japanese nasals by making use of the theory of Gradient Symbolic Representation (GSR), originally proposed by Smolensky & Goldrick (2016), and developed later by various scholars, including Rosen (2016), Zimmerman (2018), Walker (2020) and many others. The gist of this general proposal is that phonological structures and features, such as e.g., [voice], can be partially—or numerically—activated, going beyond the presence/absence dichotomy. Walker (2020) for example argues that in Korean, labials, coronals, and dorsals have activation values of 0.8, 0.7, and 0.9, respectively. The current proposal making use of this general theory, inspired in particular by Walker (2020), is as follows:

- (1) Japanese nasals are [0.6 voice].
- (2) Japanese voiced obstruents are [1 voice].
- (3) Lyman’s Law prohibits a morpheme which has [2.0 voice] feature or higher.
- (4) Post-nasal voicing can be triggered by a segment that has at least [0.5 voice].

In essence, our current proposal takes the ambivalent behavior of nasals at its face value and posits that Japanese nasals are specified for [+voice], but its activation is not as strong as [+voice] for obstruents. Any activation values between 0.5 and 0.99 would work for the case at hand, but for the sake of simple illustration, we posit that [+voice] feature of nasals has an activation value of 0.6 in Japanese phonology. The postulates in (1) and (3) are intended to account for the blockage of rendaku by two instances of nasals, but not by one instance. One voiced obstruent suffices to

trigger Lyman's Law, hence our postulate (2). The fact that Japanese nasals can trigger voicing of following obstruents is captured by our postulate (4).

THE OT TABLEAUX: The following set of OT tableaux illustrate how our proposal accounts for all the relevant patterns of Rendaku. The constraint ranking below is identical to the ranking posited by Ito & Mester (2003) in their general analysis of Rendaku. The crucial new addition is the case of blockage by two nasals. As in tableau (d), two instances of nasals amount to having a total of [1.2 voice] value, thereby triggering Lyman's Law if the second member undergoes Rendaku. As in (c) on the other hand, one instance of nasal is not enough to trigger Lyman's Law, because one nasal has only a [0.6 voice] activation. As shown in (e), the constraint that is responsible for post-nasal voicing can be formalized as an AGREE constraint which prohibits two adjacent constraints with larger than [0.5 voice] mismatch, which outranks Faith-IO.

(a) /hosi-sora/	Lyman's Law	Rendaku	Faith-IO
[hosi-sora]		*!	
→ [hosi-z _[1 voi] ora]			*
(b) /hosi-saba/	Lyman's Law	Rendaku	Faith-IO
→ [hosi-sab _[1 voi] a]		*	
[hosi-z _[1 voi] ab _[1 voi] a]	*! (total [2 voice])		*
(c) /neko-suna/	Lyman's Law	Rendaku	Faith-IO
[neko-sun _[0.6 voi] a]		*!	
→ [neko-z _[1 voi] un _[0.6 voi] a]			*
(d) /sakura-hanami/	Lyman's Law	Rendaku	Faith-IO
→ [sakura-han _[0.6 voi] am _[0.6 voi] i]		*	
[sakura-b _[1 voi] an _[0.6 voi] am _[0.6 voi] i]	*! (total [2.2 voice])		*
(e) /kam-ta/	AGREE	Faith-IO	
[kan _[0.6 voi] -t _[0 voi] a]	*! (0.6 mismatch)		
→ [kan _[0.6 voi] -d _[1 voi] a]	(0.4 mismatch)	*	

COMPARISON WITH PREVIOUS PROPOSALS: There are three major previous proposals regarding the behavior of nasals in Japanese. Ito & Mester (1986) posit that nasals are underspecified for [voice]; Mester & Ito (1989) propose that [voice] is a privative feature and sonorants do not bear that feature at all; Rice (1993) suggests that sonorants and obstruents are specified for different [voice] features. All of these proposals account for why one instance of nasal does not trigger Lyman's Law. The underspecification approach posits that nasals get specified for [+voice] at the time during the derivation when post-nasal voicing occurs. Rice's proposal posits that post-nasal voicing is triggered by sonorant-specific voicing feature, whereas Lyman's Law specifically targets obstruent-specific voicing feature. Crucially, however, none of these proposals account for why two nasals can trigger Lyman's Law. Our proposal based on the general theory of Gradient Symbolic Representation accounts for all aspects of the ambivalent behavior of [voice] feature of Japanese nasals.

SELECTED REFERENCES: [1] Kim, S. (2019) Modeling super-gang effects in MaxEnt: Nasal in Rendaku. Talk presented at NELS. [2] Ito, J. & A. Mester (1986) The phonology of voicing in Japanese: theoretical consequences for morphological accessibility. *Linguistic Inquiry* 17: 49-73. [3] Smolensky, P. & M. Goldrick (2016) Gradient Symbolic Representations in Grammar: The case of French Liaison. Manuscript. [4] Walker, R. (2020) Gradient feature activity in Korean place assimilation. *Proceedings of NELS*.